CHAPTER 1

BRICKS BEADS AND BONES

CHAPTER ESSENCE AT A GLANCE:
Master the Harappan (Indus Valley) Civilisation: mature urban architecture, citadel vs lower town, covered drainage, craft production at Chanhudaro, and theories of collapse.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Urban Architecture and Settlement Layout:** Mohenjo-Daro and Harappa featured standardized baked bricks (ratio 1:2:4) organized into a fortified western Citadel (public platforms) and an eastern Lower Town residential grid.

• **Sanitation and Advanced Drainage:** Roads and streets laid out along an approximate grid-iron pattern; domestic drains connected through exterior walls to covered street channels equipped with inspection sumps.

• **Specialized Craft and Trade Networks:** Chanhudaro specialized in bead-making, seal-carving, and shell-cutting; raw materials procured locally (carnelian from Bharuch, lapis lazuli from Shortughai) and via maritime trade with Oman (Magan) and Mesopotamia (Meluhha).

• **Decline and De-urbanization:** Mature Harappan settlements were abandoned around 1800 BCE due to environmental factors: river drying or flooding, deforestation, climate shifts, and tectonic changes rather than a single sudden invasion.

ESSENTIAL HISTORICAL CHRONOLOGIES & THEMES

$$\text{Harappan Urban Grid} \Longleftrightarrow \text{Citadel (Public/Sacred)} + \text{Lower Town (Residential Grid)}$$

Dual Settlement Zoning and Social Architecture

$$\text{Standard Brick Proportion} \Longleftrightarrow \text{Thickness} : \text{Width} : \text{Length} = 1 : 2 : 4$$

Universal Civil Engineering Metrological Standard

$$\text{Great Bath Hydrology} \Longleftrightarrow \text{Bitumen Waterproofing} + \text{Brick-on-Edge} + \text{Drain Outflow}$$

Ritual Purification Architecture on Mohenjo-Daro Citadel

$$\text{Trade Geography} \Longleftrightarrow \text{Meluhha (Harappa)} \longleftrightarrow \text{Dilmun (Bahrain)} \longleftrightarrow \text{Magan (Oman)}$$

Bronze Age Maritime Mesopotamian Trade Connections

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Harappan seals are predominantly made of steatite (a soft soapstone) and feature animal motifs alongside an undeciphered logo-syllabic script written right-to-left.

• **Alert:** The Great Bath on the Citadel of Mohenjo-Daro was made watertight by setting bricks on edge using mortar of gypsum and a coating of natural bitumen (asphalt).

• **Alert:** Never confuse Mature Harappan (2600-1900 BCE) with Early Harappan or Late Harappan pastoral successor cultures (Cemetery H).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Describe the domestic architecture and sanitation system of Mohenjo-Daro. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Grid-Iron Street System:** Roads and alleys intersected at approximate right angles, dividing the town into rectangular residential blocks. **[1 Mark]**

• **Covered Drainage:** Household wastewater flowed into street drains built of baked bricks and mortar, covered with limestone slabs that could be lifted for periodic cleaning. **[1 Mark]**

• **Courtyard and Privacy:** Houses were centered around open courtyards with private bathrooms; there were no ground-level windows facing streets, safeguarding domestic privacy. **[1 Mark]**

2. Explain the architectural features of the Great Bath on the Citadel of Mohenjo-Daro and state its probable function. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Structure:** A large rectangular tank situated in a courtyard surrounded by a corridor on all four sides. **[1 Mark]**

• **Waterproofing:** Built of baked bricks laid on edge with gypsum mortar; made completely watertight by applying a thick lining of natural bitumen (asphalt) along the floor and walls. **[1.5 Marks]**

• **Flights of Steps and Drain:** Two wide flights of stairs descended into the tank from north and south; wastewater drained through a huge corbelled brick sewer. **[1 Mark]**

• **Probable Function:** Unanimously interpreted by archaeologists as designed for special ritual bathing or sacred state purification ceremonies. **[0.5 Mark]**

3. Why is Chanhudaro considered an important craft production center of the Harappan civilization? What crafts were practiced there? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Significance:** Chanhudaro was a tiny settlement (less than 7 hectares) exclusively devoted to specialized craft production. **[1 Mark]**

• **Specialized Crafts:** Bead-making using carnelian, jasper, and steatite; shell-cutting for bangles; seal-carving; weight-making using chert; and metallic copper-bronze smelting. **[2 Marks]**

4. Discuss the archaeological evidence indicating long-distance trade between the Harappan civilization and West Asia. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Omani Copper and Jars:** Chemical analysis shows Omani copper and Harappan artifacts both contain traces of nickel; a distinctive black-clay coated Harappan jar was discovered at an Omani site. **[1.5 Marks]**

• **Mesopotamian Texts:** Mesopotamian cuneiform tablets refer to trade with 'Meluhha' (Harappa), mentioning carnelian, lapis lazuli, copper, gold, and timber. **[1.5 Marks]**

5. State two characteristics of the Harappan script. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Characteristic 1:** It is undeciphered and logo-syllabic, containing between 375 and 400 distinct signs. **[1 Mark]**

• **Characteristic 2:** Written from right to left, as evidenced by wider spacing on the right and cramping on the left margin of seals. **[1 Mark]**

6. State any two reasons given by historians for the decline of the Harappan civilization around 1800 BCE. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Reason 1:** Major ecological and climatic shifts, including deforestation and drying up of river channels (like the Ghaggar-Hakra). **[1 Mark]**

• **Reason 2:** Excessive flooding, tectonic earthquakes altering river courses, and exhaustion of surrounding agricultural lands. **[1 Mark]**